HEALTH & FITNESS CLUB MEMBERSHIP PLATFORM

Project Report | SQL Analytics, Information Systems & Power BI

Presented by

Vishal Rathour

Primary IS classification: CRM / MIS with DSS capabilities

1. Executive Summary

The Health & Fitness Club Membership Platform connects day-to-day gym operations with management reporting and analytical decision support. It organizes member, plan, payment and attendance information, then uses SQL queries and a Power BI dashboard to examine membership mix, recorded revenue, monthly recurring revenue (MRR), engagement and churn risk.

Workflow: Operational data → relational database → SQL analysis → KPIs → Power BI dashboard → management action.

The presentation describes the platform as an integrated CRM/MIS solution with DSS capabilities. Predictive churn analysis and member recommendations are future enhancements, not implemented functionality according to the supplied material.

2. Project Objectives

• Organize member profiles, membership plans, payment records and attendance.

• Validate and analyze operational data using SQL.

• Measure membership distribution and plan-level engagement.

• Report recorded revenue and calculate MRR using active memberships.

• Identify low-engagement members as potential churn-risk cases.

• Present management KPIs through a Power BI dashboard.

• Outline development, adoption, privacy, security and operational controls.

3. Information System Classification

System	Role in the platform
TPS	Captures registration, membership purchases, payments and attendance.
CRM	Centralizes member profiles, plans, payment history and relationship information.
MIS	Converts operational records into management reports, KPIs and dashboards.
DSS	Supports decisions using churn-risk, MRR, revenue trends and attendance analysis.
KMS	Organizes operational and customer knowledge for management use.
AI (future)	Potential predictive churn analysis and member recommendations.

4. System Architecture and Data Flow

The layered framework connects business processes, data, applications, technology, security and end users.

• Business: membership management, customer management, payments, attendance and reporting.

• Data: Members, Plans, Member_Plans, Payment_History and Attendance.

• Applications: CRM, MIS, DSS and Power BI dashboard.

• Technology: MySQL, Power BI, web/application layer and payment gateway.

• Security: authentication, role-based access, encryption, backup/recovery and payment security.

• Users: members, staff, managers and administrators.

The presentation includes Level 0 and Level 1 DFDs. External entities are Member, Admin/Staff, Payment Gateway and Management. Level 1 processes are Member Management, Plan Management, Payment Processing, Attendance Management, and Analytics & Reporting. Data stores are D1 Members, D2 Plans, D3 Payment_History and D4

Attendance.

5. Development Methodology: Agile SDLC

Agile was selected for iterative development, continuous feedback and flexible requirements.

Sprint	Main deliverable
1	Member and plan management
2	Payment and membership processing
3	Attendance management
4	MIS dashboard and reports
5	DSS analysis: churn risk and MRR
6	Security, testing and final improvements

Each sprint follows requirements, planning, design/development, testing, review/feedback and improvement.

6. Database Design

Table	Key fields / purpose
Members	member_id (PK), full_name, email, gender, date_of_birth, city, join_date, status
Plans	plan_id (PK), plan_name, duration_months, monthly_fee, description
Member_Plans	membership_id (PK), member_id (FK), plan_id (FK), start_date, end_date, membership_status
Payment_History	payment_id (PK), member_id (FK), plan_id (FK), payment_date, amount, payment_method, payment_status
Attendance	attendance_id (PK), member_id (FK), attendance_date, check_in_time, check_out_time

The presentation also uses a source behavioural table named gym_membership, including visit_per_week, abonoment_type and avg_time_check_in. It distinguishes this source behavioural analysis from normalized relational-table analysis.

7. SQL Analytics

7.1 Data validation

• Count records in gym_membership_raw and gym_membership.

• Inspect sample rows.

• Review plan distribution using abonoment_type.

7.2 Membership and plan analysis

Count total members, analyze gender distribution, and count members by plan by joining Member_Plans with Plans.

7.3 Revenue and payment analysis

Calculate payment count and recorded revenue by payment method and by plan, using Payment_History joined with Plans. The dashboard can show revenue trends using payment_date.

7.4 Monthly Recurring Revenue (MRR)

MRR is defined in the presentation as the sum of monthly fees for currently active memberships:

```
SELECT SUM(p.monthly_fee) AS MRR
FROM Member_Plans mp
JOIN Plans p ON mp.plan_id = p.plan_id
```

```
WHERE mp.membership_status = 'Active'
  AND CURDATE() BETWEEN mp.start_date AND mp.end_date;
```

This is distinct from historical recorded revenue, which is calculated from Payment_History.amount.

7.5 Engagement segmentation

• 0 visits/week: No Visit

• 1 visit/week: Low Engagement

• 2–3 visits/week: Moderate Engagement

• More than 3 visits/week: High Engagement

7.6 Churn-risk segmentation

• 0–1 visits/week: High Risk

• 2 visits/week: Medium Risk

• More than 2 visits/week: Low Risk

Interpretation: The presentation explicitly describes this as engagement-based churn risk. Confirmed cancellation data is not present in the source query, so this is not a verified customer cancellation rate.

8. Power BI Dashboard

The supplied presentation reports this dashboard snapshot:

KPI	Reported value
Total Members	1,000
Total Recorded Revenue	3.99M
Premium Members	493
Standard Members	507
High-Risk Members	194

Dashboard sections shown include executive KPI cards, membership plan mix, revenue by membership plan, gender distribution, engagement analysis, potential churn-risk analysis and revenue by payment method.

These values and views are reproduced from the supplied presentation; they have not been independently recalculated in this report.

9. Technology Adoption and Business Process

The adoption journey is Awareness → Interest → Evaluation → Trial → Adoption → Regular Use.

• Members: registration, plan selection, online payment and attendance.

• Admin/Staff: member and plan management, payment verification and attendance.

• Management: dashboard, revenue, churn risk, MRR and attendance analytics.

End-to-end process: member registration → plan selection → e-payment → payment verification → membership activation → attendance tracking → dashboard and reports.

10. E-Payment Integration

• Member selects a membership plan and initiates payment.

• The platform sends a payment request to the gateway.

• The gateway returns transaction status; the platform verifies it.

• Verified transaction data is stored in Payment_History.

• Successful payment triggers membership confirmation and receipt/status.

Payment status should be verified before activating membership or treating a payment record as successful.

11. Security, Privacy and Cybercrime Mitigation

• Authentication and strong password practices.

• Role-Based Access Control for Members, Admin/Staff and Management.

• Encryption for information in storage and transmission.

• Secure payment gateway communication and transaction verification.

• Restricted database access, controlled backup and recovery.

• Audit logs for important login, payment and administrative actions.

• Phishing awareness and incident response: identify → contain → report → recover → review.

Privacy principles include data minimization, authorized access, legitimate and transparent use, consent, avoiding unnecessary sensitive payment details, and controlled backups. Several compliance actions are marked Planned in the presentation; they are recommendations, not independently audited or certified implementations.

12. Environmental and Ergonomic Audit

• Comfortable chair, desk and workstation posture.

• Appropriate monitor height and viewing distance.

• Adequate lighting and ventilation.

• Safe electrical sockets, cables and connections.

• Energy-efficient use of IT equipment.

• Responsible segregation and recycling/disposal of e-waste.

13. Quality-Control Considerations

• Avoid duplicate member-loading INSERT statements.

• Do not add a second primary key to Member_Plans if membership_id is already its primary key.

• Keep MRR logic separate from historical payment totals.

• Label engagement-based churn risk accurately.

• Use normalized tables for relational KPIs and gym_membership for source behavioural analysis.

14. Conclusion

The platform demonstrates how operational membership data can be organized and converted into management insights. SQL analysis and the Power BI dashboard bring together member and plan distribution, recorded revenue, MRR, engagement and potential churn-risk indicators. The project also documents an Agile development approach and outlines payment, security, privacy, ethics, environmental and ergonomic considerations.

Its principal value is connecting operational records with data-informed management decisions.

15. Future Scope

• Capture explicit membership cancellation and renewal outcomes to calculate actual churn.

- Add validated historical snapshots for month-over-month KPI comparisons.
- Introduce predictive churn modelling after collecting sufficient quality data.
- Add automated alerts for expiring memberships and low-engagement members.
- Extend member self-service for profile, plan and payment history.
- Strengthen security testing, backup validation and audit-log review.

Tools: MySQL / SQL, Microsoft Power BI, relational database concepts, Data Flow Diagrams and Agile SDLC.